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CLASS : CSAI-D

ROLL NO: 34

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```
import pandas as pd
import numpy as np
```

```
# Read CSV file
df_csv = pd.read_csv("/Exam_Score_Prediction.csv")


df_csv.head()
```

	student_id	age	gender	course	study_hours	class_attendance	internet_access	sleep_hours	sleep_quality	study_method
0	1	17	male	diploma	2.78	92.9	yes	7.4	poor	coaching
1	2	23	other	bca	3.37	64.8	yes	4.6	average	online videos
2	3	22	male	b.sc	7.88	76.8	yes	8.5	poor	coaching
3	4	20	other	diploma	0.67	48.4	yes	5.8	average	online videos
4	5	20	female	diploma	0.89	71.6	yes	9.8	poor	coaching

New section

```
#shape of data
df_csv.shape
```

```
(20000, 13)
```

```
#missing values
df_csv.isnull().sum()
```

	0
student_id	0
age	0
gender	0
course	0
study_hours	0
class_attendance	0
internet_access	0
sleep_hours	0
sleep_quality	0
study_method	0
facility_rating	0
exam_difficulty	0
exam_score	0

dtype: int64

```
#datatypes of each column
df_csv.dtypes
```

	0
student_id	int64
age	int64
gender	object
course	object
study_hours	float64
class_attendance	float64
internet_access	object
sleep_hours	float64
sleep_quality	object
study_method	object
facility_rating	object
exam_difficulty	object
exam_score	float64

dtype: object

```
df_csv.rename(columns={"exam_score": "Marks"}, inplace=True)
```

```
#finding zero values
(df_csv == 0).sum()
```

	0
student_id	0
age	0
gender	0
course	0
study_hours	0
class_attendance	0
internet_access	0
sleep_hours	0
sleep_quality	0
study_method	0
facility_rating	0
exam_difficulty	0
exam_score	0

dtype: int64

```
#selecting columns
#df_csv["study_hours"]

df_csv[["student_id", "class_attendance", "study_hours"]]
```

Show hidden output

```
#sorting
df_csv.sort_values("class_attendance", ascending=False)
```

	student_id	age	gender	course	study_hours	class_attendance	internet_access	sleep_hours	sleep_quality	study_me
12998	12999	18	female	b.sc	5.90	99.4	yes	8.6	poor	coa
18575	18576	21	female	bba	6.73	99.4	yes	6.7	poor	n
2466	2467	19	other	b.tech	5.49	99.4	yes	4.5	good	online vi
11263	11264	18	female	diploma	0.19	99.4	yes	9.7	good	n
11257	11258	17	other	b.tech	6.25	99.4	yes	5.2	average	self-
...	...	...	...	...	...	...	...	...	...	
11910	11911	18	male	b.com	2.87	40.6	yes	9.9	poor	self-
17749	17750	17	male	b.tech	3.78	40.6	yes	9.6	poor	online vi
17795	17796	19	other	b.tech	7.54	40.6	yes	4.3	good	group
5170	5171	21	male	b.com	2.82	40.6	yes	6.2	good	online vi
25	26	22	male	b.com	7.91	40.6	yes	4.3	good	group

20000 rows × 13 columns

Start coding or generate with AI.

```
#counting unique value
df_csv["gender"].value_counts()
```

gender	count
other	6726
male	6695
female	6579

dtype: int64

```
#converting datatypes
df_csv["age"] = df_csv["age"].astype(float)
df_csv["age"].dtype
```

```
dtype('float64')
```

```
#grouped students mean exam score on the basis  of course
df_csv.groupby("course")["exam_score"].mean()
```

course	exam_score
b.com	62.333197
b.sc	62.718993
b.tech	62.653529
ba	62.112214
bba	62.928127
bca	62.524078
diploma	62.330633

dtype: float64

```
import pandas as pd

data = {
    "student_id": [1, 2, 3, 4],
    "age": [20, 21, 22, 20],
    "gender": ["Male", "Female", "Male", "Female"],
    "exam_score": [85, 72, 90, 60]
}

df = pd.DataFrame(data)
```

```
df
```

	student_id	age	gender	exam_score
0	1	20	Male	85
1	2	21	Female	72
2	3	22	Male	90
3	4	20	Female	60

```
df.iloc[0:3]
```

	student_id	age	gender	exam_score
0	1	20	Male	85
1	2	21	Female	72
2	3	22	Male	90